

TrustVote

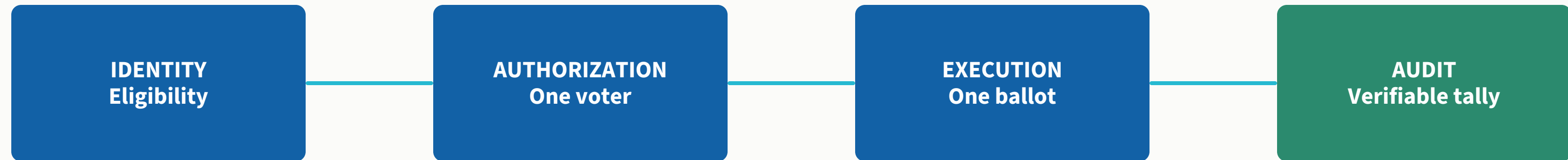
A Decentralized Voting System on Blockchain

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Introduction

Blockchain provides a shared, append-only execution layer for elections where integrity must be independently verifiable.



Trust is shifted from a single administrator to transparent rules, replicated state, and inspectable evidence.

Problem Statement

Conventional digital voting concentrates critical trust in infrastructure that voters and observers cannot independently inspect.

01

CENTRAL CONTROL

A privileged operator can alter configuration, records, or availability.

02

WEAK AUDITABILITY

Database logs are controlled by the same administrative boundary.

03

VERIFIABILITY GAP

A voter cannot prove inclusion without exposing ballot secrecy.

Objectives

Define a secure, auditable prototype and evaluate its behavior under functional and adversarial conditions.

01

Tamper-evident election state

02

Role-bound administration

03

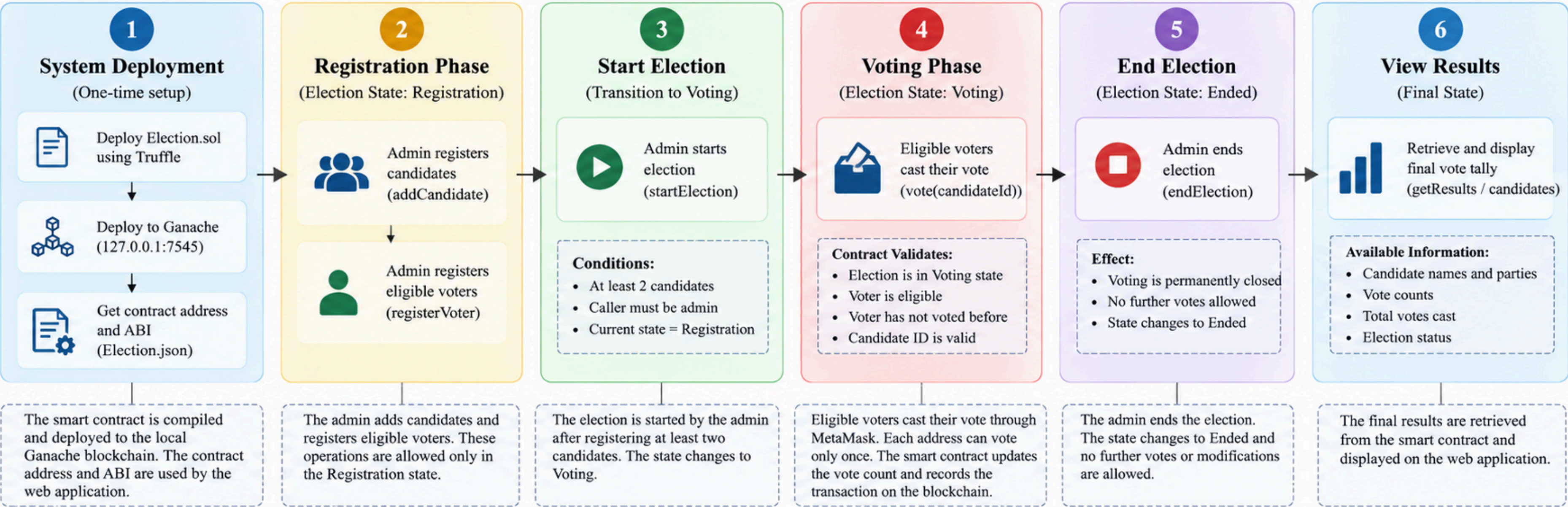
One-voter / one-ballot enforcement

04

Reproducible validation evidence

Election Lifecycle in TrustVote

From Setup to Final Results



Election State Transition

Registration
(Candidates and voters can be registered by admin)

startElection()
(Admin only)

Voting
(Eligible voters can cast votes)

endElection()
(Admin only)

Ended
(No further votes allowed)

Limitations of Existing System

The design baseline is a centrally hosted web application backed by a mutable database.

CENTRALIZED BASELINE

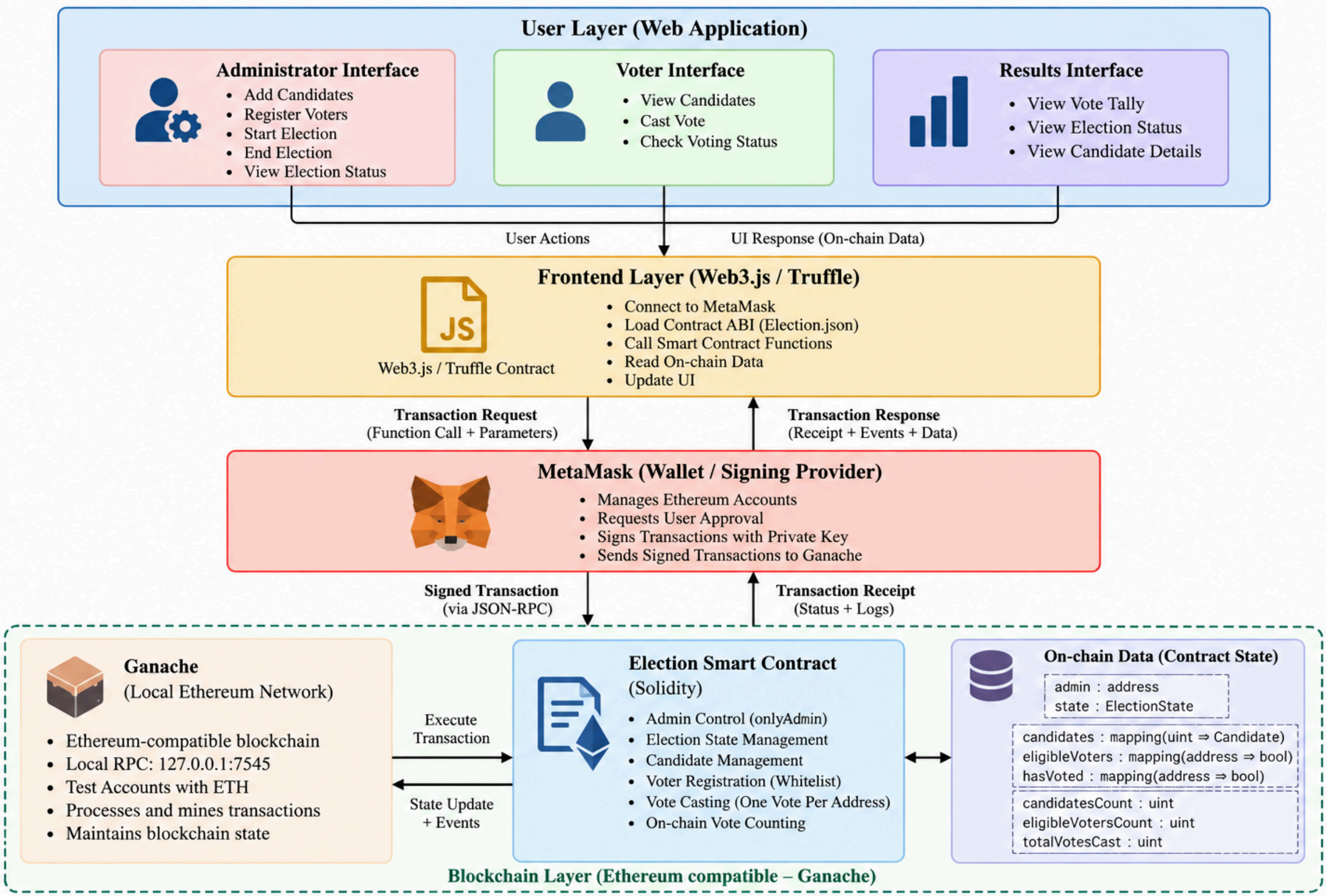
- Single database authority
- Administrator-visible write path
- Opaque recount procedure
- Audit after the event

TRUSTVOTE RESPONSE

- Replicated append-only ledger
- Contract-enforced state transitions
- Deterministic public tally
- Continuous transaction evidence

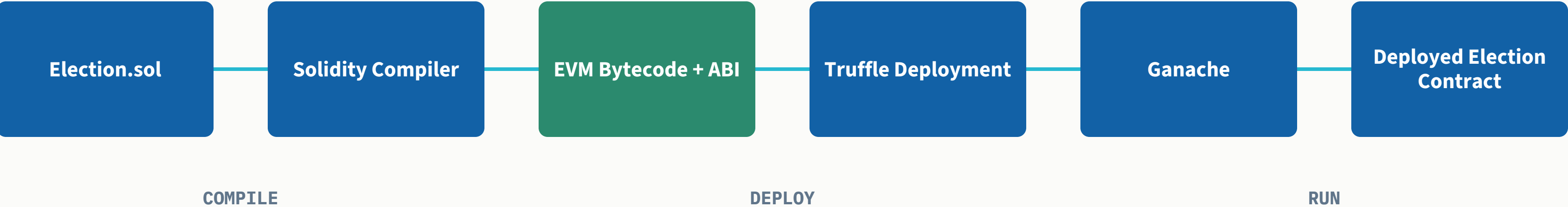
Proposed Architecture

The browser connects to the blockchain via Web3.js and MetaMask, submitting signed transactions that the Election smart contract validates and records on the Ganache blockchain.



Technology Stack & Development Pipeline

Election.sol is compiled to EVM bytecode and deployed via Truffle onto a local Ganache blockchain.

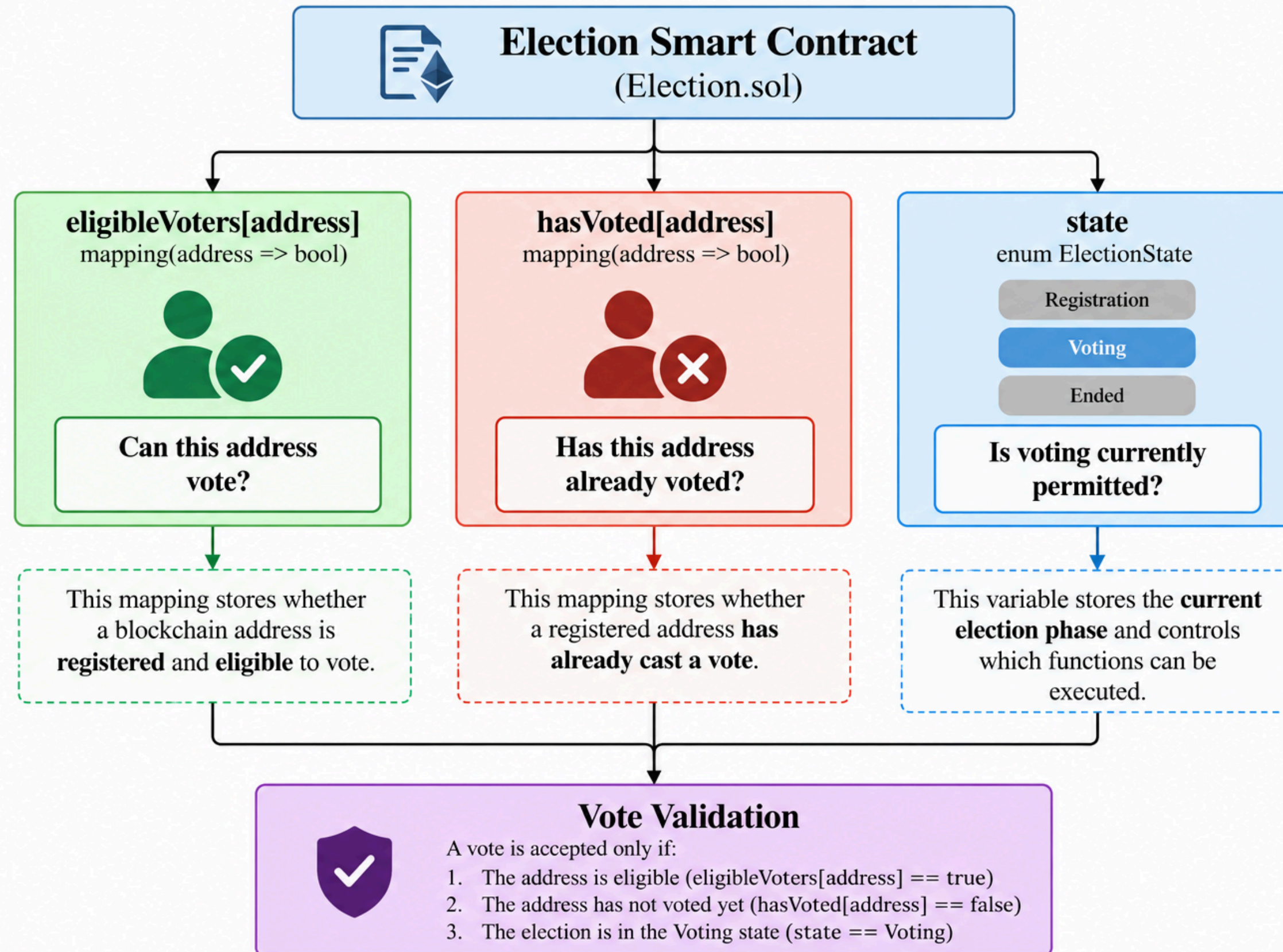


Solidity • Web3.js • MetaMask • Truffle • Ganache

*EVM bytecode is the low-level, hexadecimal machine language that the Ethereum Virtual Machine (EVM) executes on the blockchain.

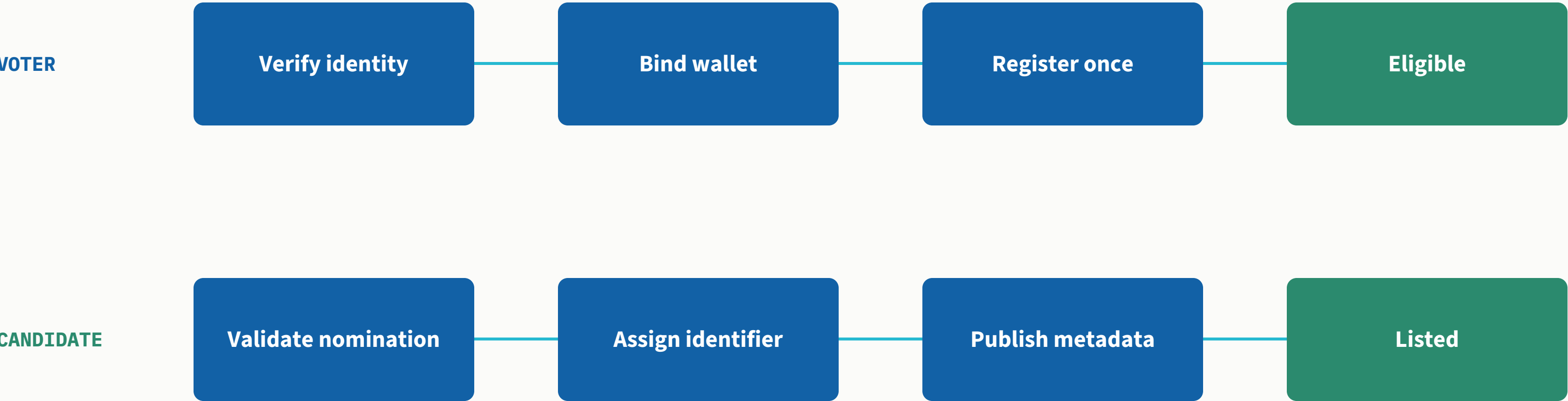
Core Design of Smart Contract

State and authorization are explicit; invalid transitions revert before ledger state changes.



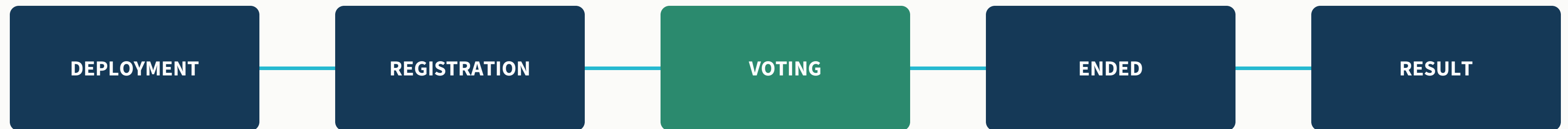
Voter Registration & Candidate Management

Administrative actions are separated from voter actions and remain attributable on-chain.



Election Lifecycle & Voting Algorithm

Contract enum: Registration → Voting → Ended (one-directional, irreversible).



`vote(candidateId)`

```
1  require state == Voting
2  require eligibleVoters[voter] && !hasVoted[voter]
3  require valid candidateId
4  mark voted; increment vote count; emit VoteCast
```

**ATOMIC
STATE CHANGE**
VALIDATE → WRITE → EMIT

Security Mechanisms

Controls are layered across identity, contract execution, ledger integrity, and operational evidence.

AUDITABILITY • immutable events, transaction receipts, reproducible logs

LEDGER INTEGRITY • replicated state, ordered commits, tamper evidence

CONTRACT SAFETY • role checks, phase guards, revert-on-failure

IDENTITY • eligibility binding, unique wallet, least privilege

Threats addressed: unauthorized administration • duplicate voting • invalid candidate • out-of-phase transaction • record tampering

Limitations & Future Scope

The prototype establishes integrity controls; deployment-grade privacy and governance require further work.

CURRENT LIMITS

- 1 Prototype-scale network
- 2 Wallet–identity link is permissioned
- 3 Ballot secrecy not cryptographically complete
- 4 Performance not tested at national scale

NEXT RESEARCH

- 1 Zero-knowledge eligibility proofs
- 2 Threshold / homomorphic tally
- 3 Formal verification and fuzzing
- 4 Multi-site deployment and load study

Conclusion

TrustVote demonstrates how election rules can be made explicit, executable, and independently auditable.

INTEGRITY
Append-only evidence

ENFORCEMENT
Contract-bound lifecycle

ASSURANCE
Repeatable test artefacts

The contribution is a defensible prototype architecture—not a claim that blockchain alone solves electoral trust.

References

References cited in the TrustVote dissertation.

[1] M.-V. Vladucu, Z. Dong, J. Medina, and R. Rojas-Cessa, “E-Voting Meets Blockchain: A Survey,” IEEE Access, vol. 11, pp. 23293–23308, 2023, doi: 10.1109/ACCESS.2023.3253682.

[2] M. Hajian Berenjestanaki, H. R. Barzegar, N. El Ioini, and C. Pahl, “Blockchain-Based E-Voting Systems: A Technology Review,” Electronics, vol. 13, no. 1, art. 17, 2024, doi: 10.3390/electronics13010017.

Thank you • Questions